How exactly do we use our current state information to decide what action to take? This is where the key concepts of *value functions* and *policy functions* come into play, which we already have a bit of experience with. Let's first tackle policies.

In words, a policy, π, is the strategy of an agent in some environment. For example, the strategy of the dealer in Blackjack is to always hit until they reach a card value of 17 or greater. It's a simple fixed strategy. In our *n*-armed bandit problem, our policy was an epsilon-greedy strategy. In general, a policy is a function that maps a state to a probability distribution over the set of possible actions in that state.

**Table 2.5. The policy function**

| Math | English |
|---|---|
| $\pi; s \rightarrow Pr(A \mid s)$, where $s \in S$ | A policy, π, is a mapping from states to the (probabilistically) best actions for those states. |

In the mathematical notation, $s$ is a state and $Pr(A \mid s)$ is a probability distribution over the set of actions $A$, given state $s$. The probability of each action in the distribution is the probability that the action will produce the greatest reward.

## 2.7.2. Optimal policy

The policy is the part of our reinforcement learning algorithm that chooses actions given its current state. We can then formulate the *optimal policy*—it's the strategy that maximizes rewards.

**Table 2.6. The optimal policy**

| Math | English |
|---|---|
| $\pi^* = \arg max\ E(R \mid \pi)$, | If we know the expected rewards for following any possible policy, $\pi$, the optimal policy, $\pi^*$, is a policy that, when followed, produces the maximum possible rewards. |